WeanAge

Open-source weaning-age estimation for livestock from incremental dentine $\delta^{15}N$

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What it is. WeanAge turns a series of $\delta^{15}N$ values measured along a tooth into a weaning age with an honest uncertainty, for cattle, pigs and horses (and any species with a tooth growth table). It compares several curve models, corrects for the blurring caused by slicing, compares sites and species, and helps plan sampling before any tooth is cut. It runs in a web browser with nothing to install, or as a local app and command line.

Why it matters. When animals were weaned shows how herds were managed. Early weaning frees milk for people, staged weaning points to dairying, and the timing reflects seasonal births (Balasse & Tresset 2002; Balasse et al. 2021). Tools so far were built for humans (WARN: Tsutaya & Yoneda 2013; WEAN: Ganiatsou et al. 2023), so each livestock study has written its own model, and the results are hard to compare. WeanAge aims to be one documented, tested method that any lab can use.

The principle

A suckling animal feeds one step up the food chain from its mother, so its tissues carry about 3–4‰ more ^{15}N (DeNiro & Epstein 1981; Minagawa & Wada 1984). After weaning, $\delta^{15}N$ falls to the adult level. Dentine grows in layers and is not remodelled, so thin slices cut along a tooth record this fall month by month (Balasse et al. 2001; Beaumont et al. 2013).



A synthetic cattle M2 in WeanAge. Dots are dentine slices (the hollow point failed collagen QC and is not fitted). Two of the five models are shown: logistic (blue, solid) and two-phase (orange, dash-dot). Dashed vertical lines mark each model's weaning completion with its shaded 95 % interval; the top axis gives the position on the tooth. The two-phase completion falls after the last slice, so the app flags it as an extrapolation: the logistic estimate (9.3 months) is the better supported one here.

Methods, and why they are built this way

Step	What WeanAge does	Why	Based on
Position → age	Converts each slice's distance from the cusp to an age at formation, using a growth table for the species and tooth, or a calibration from known-age teeth. Teeth of one animal are joined into one timeline.	Weaning is an age, but the lab measures millimetres. Errors in the growth table move every slice of a tooth together, so they are added as one shared shift.	Cattle crown chronology: Brown et al. 1960
Collagen quality	Flags slices by C:N, %C, %N and yield; archaeological and modern presets.	Degraded or contaminated collagen gives false $\delta^{15}N$ values. Modern collagen needs a tighter C:N window.	DeNiro 1985; van Klinken 1999; Guiry & Szpak 2020
Five curve models	Logistic, piecewise linear, exponential, two-phase (staged weaning) and a WEAN-style polynomial "elbow".	The shape of weaning is not known in advance, and the chosen shape changes the age. Fitting all of them shows how much the answer depends on the choice. Two-phase captures supplementary feeding before final weaning.	Tsutaya & Yoneda 2013; Ganiatsou et al. 2023
Model comparison	AICc, Akaike weights, and a model-averaged weaning age.	AICc balances fit against complexity in small samples; averaging avoids betting on one model.	Hurvich & Tsai 1989; Burnham & Anderson 2002
Uncertainty	95 % intervals from the fit's covariance or a residual bootstrap, plus the age-model error.	Every weaning age should come with how sure we can be.	Efron & Tibshirani 1993
Slice blurring	Optionally fits each slice as the average of the curve over its height and the dentine depth, with oblique growth layers and less volume towards the pulp.	Each slice mixes weeks of growth, which smooths and delays the signal. Modelling the mixing recovers the real diet change.	Tsutaya 2020 (MDSS)
Plausibility checks	Warns when the drop is below the noise or above one food-chain step; when $\delta^{15}N$ rises with age; when there is a yearly cycle; and when $\delta^{13}C$ falls across weaning in cattle.	Each points to something other than weaning: noise, stress, seasonal diet or mixed samples. In known-age calves $\delta^{13}C$ rose as $\delta^{15}N$ fell.	Fuller et al. 2006; Beaumont & Montgomery 2016; Balasse et al. 2001, 2021

Step	What WeanAge does	Why	Based on
Comparing groups	Random-effects means, permutation and Holm-corrected tests, and hierarchical Bayesian models: two-stage (pooling the estimates) or one-stage (fitting every tooth's raw values together).	Animals genuinely differ, and each estimate has its own uncertainty. Bayesian models give direct answers such as "probability pigs were weaned earlier than cattle". The one-stage model carries each curve's uncertainty all the way through.	DerSimonian & Laird 1986; Holm 1979; Gelman 2006; Vehtari et al. 2021
Validation	Bias, error, correlation and interval coverage against known-age animals or published estimates.	A method is only as good as its record on animals whose weaning age is known.	Balasse et al. 2001; Ganiatsou et al. 2023
Sampling planner	Simulates teeth with known weaning (of several curve shapes), samples them as a protocol would, and scores each model.	Shows the slice size, number of teeth and precision needed before material is destroyed.	—

Design principles

- **Transparent about what is not yet known.** Every value still awaiting lab work (growth tables, layer geometry, precision) is marked `TBD-LAB` and highlighted in the app and in each result, so placeholder results cannot pass as real.
- **Reproducible.** Projects save to one file. Reports record every setting and source, and known-answer tests keep results stable between versions.
- **Accessible.** Plain-language help for every term, a high-contrast dark theme with a light mode (WCAG AAA text), colour-blind-safe curves with distinct line patterns, and full keyboard use.
- **Nothing to install.** The browser version runs the same Python code inside the page (Pyodide). Data never leaves the computer, and it works offline after the first visit.

Status and how to take part

WeanAge is a working draft. The method runs end to end, but **the tooth growth tables and several lab parameters are placeholders**, so results are illustrative until they are measured. Contributions that would help most:

- known-age reference animals (dentine $\delta^{15}N$ series with recorded weaning dates),
- tooth growth data for pigs and horses, and crown heights for cattle,
- testing on published series, and feedback on the interface.

Code, guide and documentation: https://github.com/andymaehartley/repotest
Cite: Hartley, A. WeanAge: weaning-age estimation from incremental dentine $\delta^{15}N$ (version 0.2). See `CITATION.cff` in the repository.

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